

TEAM ASSIGNMENT #2

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
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Due Date: Friday September 20 by 5:00pm in my office

Number of points.....100

Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

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1. Executive Summary

The goal of this assignment was to design an excellent production plant for the production of cyclohexane with highest purity. We plan to design this process by hydrogenating benzene in a bubble column reactor. To approach the above said goal, material balance for each component was planned out using mass conservation concept. Also, mass balance was carried out for each equipment to calculate the molar flow rates of the reactants and the products. Using the given physical properties, and some flow rates, a flowsheet for the convergence of the process was prepared to assure the given conversion and selectivity. The excel calculator will be utilized to and to get an estimate of product produced, its flow rate and its purity. The purity of cyclohexane should meet the customer's requirements and market specifications. In this report we also provide a good estimate of Fixed Capital investment for a 200KMTA world class manufacturing plant in 2016 dollars. After analyzing the data for 2016 inflation rate[1], Fixed Capital investment for a 200KMTA was found to be 142 M dollars.

2. Systems & Procedures (components, physical properties used, equipment)

The benzene feed for this problem was available at $T = 100^{\circ}\text{F}$ and $P = 15$ psia with a flow rate of 100 lbmol/hr. Hydrogen was available at $T = 120^{\circ}\text{F}$ and $P = 335$ psia with a flowrate of 313 lbmol/hr. The purity of the Hydrogen in the feed is 97.5 % along with the methane that makes 2.0 % and nitrogen that makes 0.5 % of this feed. Both the feed are mixed and fed to a bubble - column reactor with catalyst of nickel. The properties of the reactor is given as, $T = 400^{\circ}\text{F}$ and $P =$ psia. The effluent from the reactor is cooled and flashed at $T = 120^{\circ}\text{F}$ and $P = 310$ psia to give hydrogen rich gas and liquid mix stream of benzene, hydrogen, nitrogen, methane and cyclohexane. The hydrogen gas stream is recycled back along with the liquid stream to convert the unreacted benzene. The hydrogen gas recycle stream is modeled to maintain hydrogen to benzene ratio to 3.3. The liquid recycle stream is limited to no more than 45% of fresh liquid benzene feed. The remaining liquid is flash again in a low-pressure separator to give out the desired product of cyclohexane and fuel gas from the top of the separating column.

3. Results:

The mass balance flow sheet converged after 185 iterations to give overall benzene conversion of 99.50% and selectivity of 99.70%. The Product Flow rate for the given feed flow rates after the iterations was obtained to be 270.28 KMTA. The Fixed Capital Investment, MM USD was estimated to be 173.08.

The results of the mass balance, overall conversion, selectivity, final feed flow rates and the product purity is shown below.

Table 1: Estimation of Overall Conversion and Selectivity

Overall Benzene Conversion, mol%	99.50
Overall Benzene Selectivity, mol %	99.70

Table 2: Estimation of mass flow rate of the final product and its purity

<u>Stream Components & Properties</u>	<u>S10 (Product)</u>
hydrogen, lbmol/hr	0.002728
nitrogen lbmol/hr	0.006817
methane, lbmol/hr	0.185934
benzene, lbmol/hr	0.338292
cyclohexane, lbmol/hr	97.653108
Total Molar Flow Rate, lbmol/hr	98.186880
<u>Product Purity, wt%</u>	99.64

Comment whether the purity meets market specs

In most markets, cyclohexane is produced at the minimum purity of 99.9%. Cyclohexane is a very useful product and is used in the production of intermediates for nylon. Nylon is in turn have many uses in clothing, thermoplastics, and carpets. Cyclohexane is also used as a solvent in industries and recently being used as a substitute for benzene [2]

4. Conclusion

In this assignment, we were able to design the best production plant for the production of cyclohexane. The purity of the cyclohexane was found to be 99.5% which is very high since we used the bubble column reactor to achieve the said goal. The mass balance that was carried out for each component across the equipment gave us the flow rates of the species involved in the process. Using these discovered flow rates through manipulating the VBA code in excel calculator, we were able to find the convergence of the flow sheet and gave us the good estimate of overall benzene conversion and selectivity. We also reported the FCI of this process to be equal to 142 M dollars for the 200 KMTA production per year.

5. Appendices

I. Appendix 1

$$n_{C_6H_6} = n_{C_6H_6}^0 - \xi_1 - \xi_2$$

$$n_{H_2} = n_{H_2}^0 - 3 \xi_1 - 9 \xi_2$$

$$n_{N_2} = n_{N_2}^0$$

$$n_{C_6H_{12}} = n_{C_6H_{12}}^0 + \xi_1$$

$$n_{CH_4} = n_{CH_4}^0 + 6 \xi_2$$

$$FCI = N (F)^{0.65}$$

$$Selectivity = \frac{\xi_1}{\xi_1 + \xi_2}$$

$$Conversion = \frac{\xi_1 + \xi_2}{n_{C_6H_6}^0}$$

$$RR - Equation = \sum \frac{z_i (k_i - 1)}{1 + a(k_i - 1)} = 0$$

VBA Modified Code:

'Specify the Fresh feed and set the tear stream

'-----

ncc = Range("comps").Count

ncc1 = ncc + 4

For i = 1 To ncc

s0H(i) = Me.Cells(8 + i, 2)

s0B(i) = Me.Cells(8 + i, 3)

tear(i) = s0H(i) + s0B(i)

Next i

'Begin the Direct Substitution Loop

'-----

done = DirectSubCH(conv, sel, gaspurge, liqRecycle, eps, q, maxite, ite)

'Populate the Spreadsheet with the iteration results

'-----

For i = 1 To ncc

Me.Cells(8 + i, 4) = s1(i)

Me.Cells(8 + i, 5) = tear(i)

Next i

Me.Cells(8 + ncc1, 4) = WorksheetFunction.Sum(s1)

Me.Cells(8 + ncc1, 5) = WorksheetFunction.Sum(tear)

'Material Balance Check

'-----

Mw(1) = 2 * H

Mw(2) = 2 * N

Mw(3) = C + 4 * H

Mw(4) = 6 * C + 6 * H

Mw(5) = 6 * C + 12 * H

'Solve for the reactor

'-----

ksi1 = tear(4) * conv * sel

ksi2 = conv * tear(4) * (1 - sel)

s2(1) = tear(1) - 3 * ksi1 - 9 * ksi2 'hydrogen

s2(2) = tear(2) 'nitrogen

s2(3) = tear(3) + 6 * ksi2 'methane

s2(4) = tear(4) - ksi1 - ksi2 'benzene

s2(5) = tear(5) + ksi1 'cyclohexane

'Solve for the high pressure separator

'-----

Kv(1) = 114.379 'Hydrogen

Kv(2) = 41.0847 'Nitrogen

Kv(3) = 15.5852 'Methane

Kv(4) = 0.0220743 'Benzene

Kv(5) = 0.020994 'cyclohexane

For i = 1 To ncc

zF(i) = s2(i) / WorksheetFunction.Sum(s2)

Next i

Call VtoF(zF, Kv, eps, maxite, alpha)

s3(1) = s2(1) * (1 - alpha) / (1 + alpha * (Kv(1) - 1))

s3(2) = s2(2) * (1 - alpha) / (1 + alpha * (Kv(2) - 1))

s3(3) = s2(3) * (1 - alpha) / (1 + alpha * (Kv(3) - 1))

s3(4) = s2(4) * (1 - alpha) / (1 + alpha * (Kv(4) - 1))

s3(5) = s2(5) * (1 - alpha) / (1 + alpha * (Kv(5) - 1))

For i = 1 To ncc

s4(i) = s2(i) - s3(i)

Next i

'Solve for the overhead splitter

'-----

For i = 1 To ncc

s6(i) = gaspurge * s4(i)

s5(i) = (1# - gaspurge) * s4(i)

```

Next i

'Solve for bottom splitter
'-----
For i = 1 To ncc
s7(i) = liqRecycle * s3(i)
s8(i) = (1# - liqRecycle) * s3(i)
Next i

'Solve for the mixer
'-----
For i = 1 To ncc
s1(i) = s0B(i) + s0H(i) + s5(i) + s7(i)
Next i

'Calculate the error and do convergence test
'-----
For i = 1 To ncc

delta(i) = Abs(s1(i) - tear(i))
If (delta(i) / WorksheetFunction.Max(1#, delta(i)) > eps) Then DirectSubCH = False
Next i

'Update the tear stream
'-----
If (DirectSubCH Or maxite = 1) Then Exit For
If (K < maxite) Then
For i = 1 To ncc
If (K <= 5) Then tear(i) = s1(i)
If (K > 2) Then tear(i) = q * tear(i) + (1 - q) * s1(i)
Next i
End If
Next K
ite = K
If K > maxite Then ite = K - 1

'Solve for the low-pressure separator
'-----
Kv(1) = 2464.97 'Hydrogen
Kv(2) = 840.15 'Nitrogen
Kv(3) = 305.311 'Methane
Kv(4) = 0.315599 'Benzene
Kv(5) = 0.295541 'cyclohexane

For i = 1 To ncc
zF(i) = s8(i) / WorksheetFunction.Sum(s8)
Next i

```

Call VtoF(zF, Kv, eps, maxite, alpha)

```
s10(1) = s8(1) * (1 - alpha) / (1 + alpha * (Kv(1) - 1))  
s10(2) = s8(2) * (1 - alpha) / (1 + alpha * (Kv(2) - 1))  
s10(3) = s8(3) * (1 - alpha) / (1 + alpha * (Kv(3) - 1))  
s10(4) = s8(4) * (1 - alpha) / (1 + alpha * (Kv(4) - 1))  
s10(5) = s8(5) * (1 - alpha) / (1 + alpha * (Kv(5) - 1))
```

```
For i = 1 To ncc  
s9(i) = s8(i) - s10(i)  
Next i
```

End Function

Sub VtoF(zi() As Double, Ki() As Double, eps As Double, maxite As Integer, alpha As Variant)

'Solve the Rachfor-Rice Equation Using Newton Method

'-----

Dim Aold As Double, Anew As Double

f = 0#

df = 0#

Aold = 0.01

For i = 1 To 100

For j = 1 To ncc: f = f + (zi(j) * (Ki(j) - 1)) / (1 + Aold * (Ki(j) - 1)): Next j

For j = 1 To ncc: df = df + (-zi(j) * (Ki(j) - 1) ^ 2) / (1 + Aold * (Ki(j) - 1)) ^ 2: Next j

Anew = Aold - f / df

f = 0: df = 0

If Abs(Aold - Anew) < eps Then Exit For

Aold = Anew

Next i

If Anew < 0# Or Anew > 1# Then

alpha = -100

Else

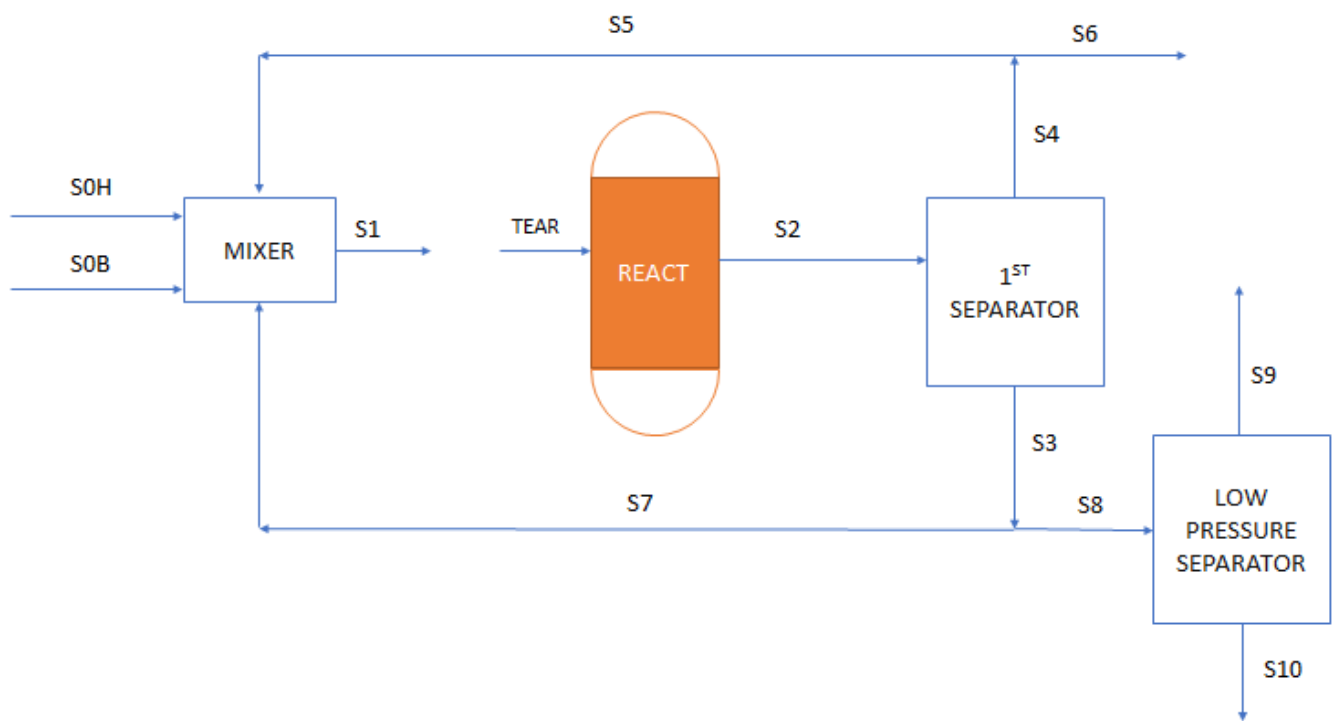
alpha = Anew

End If

If (i > 100) Then alpha = -100

End Sub

II. Appendix 2 Process Flow Diagram



6. References

1. Zhang et al, "Estimate the Capital Cost of Shale-Gas Monetization Projects," 2017.
2. Chevron Phillips Chemical Company. "Cyclohexane." *Aromatics*.